BRAC University

Department of Electrical and Electronic Engineering

EEE241/ECE241: Electromagnetic Waves and Fields

Assignment-1

Spring-2026 **Full Marks 50**

Question No. 1 **5**

State **Gauss's law**. Using Gauss's law, derive ***Maxwell's first equation*** of electromagnetics.

Question No. 2 **10**

What is **Electric Potential**? Prove that, the potential difference between two points A and B, denoted by V_{AB} in an electric field **E** caused by a point charge Q is given by

$$\frac{Q}{4\pi\varepsilon_o}\left[\frac{1}{r_B}-\frac{1}{r_A}\right]$$

Where, r_A and r_B are the magnitude of the position vectors of point A and B.

Question No. 3 **5**

Show that electric field **E** is conservative. Using the conservative nature of the electric field, prove ***Maxwell's second equation*** of electromagnetics.

Question No. 4 **10**

What is an **Electric Dipole**? Show that for an electric dipole, the Electric Potential and Field at point P is given by the equations below. Here, symbols have their usual meaning.

$$V=\frac{\mathbf{p}\cdot\mathbf{a}_r}{4\pi\varepsilon_o r^2} \qquad \mathbf{E}=\frac{p}{4\pi\varepsilon_o \mathbf{r}^3}\left(2\cos\theta\,\mathbf{a}_r+\sin\theta\,\mathbf{a}_\theta\right)$$

Question No. 5 **5**

What is a **Conductor**? Show that an isolated conductor cannot have an electric field within it.

Question No. 6 **10**

Using ohm's law, prove that, for a conductor of **uniform cross section** S, material conductivity σ and length of ℓ has a **Resistance** of

$$R = \frac{V}{I} = \frac{\ell}{\sigma S}$$

Question No. 7 **5**

What is a **Dielectric**? Write Definitions of

(i) Polar Dielectric (ii) Nonpolar Dielectric

(iii) Linear Dielectric (iv) Isotropic Dielectric (v) Homogeneous Dielectric

(vi) Dielectric Strength (v) Dielectric Breakdown

Instructions

- Assignment should be ***Hand Written.***
- Submit ***scanned pdf*** in the assignment portal.
- Use a cover page and write your ***Name*** and ***Student ID*** clearly.

Assignment-1

EEE241

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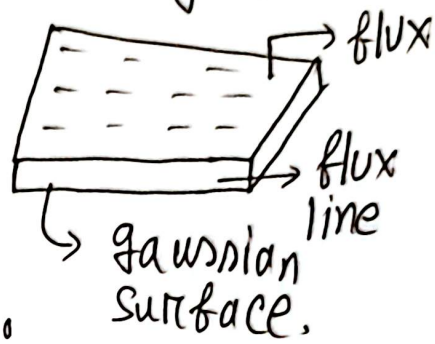
Section: 04

Ans. to the Q. NO-1

Gauss's law states that the total electric flux Passing through any closed surface is equal to the total net charge ($Q_{enclosed}$) enclosed within that surface divided by the permittivity of the free space.

$$\oint_S E\,dA = \frac{Q_{enc}}{\varepsilon_0}$$



Derivation of maxwell's 1st equation:

$$\Psi = Q_{enc}$$

$$\Psi = \oint_V d\Psi = \oint_S D \cdot ds$$

$$= \text{total charge enclosed}, Q = \int_V \rho_V \, dv$$

$$\therefore Q = \oint_S D \cdot ds = \int_V \rho_V \cdot dv$$

~~By the~~ Applying divergence,

$$\oint_S D \cdot ds = \oint_V \nabla \cdot D \cdot dv$$

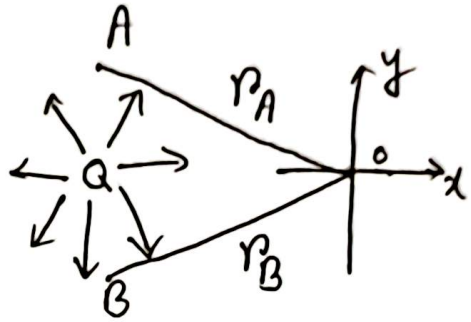
$$\boxed{\therefore \rho_V = \nabla \cdot D}$$

[Proved]

Ans. to the Q. No- 2

Electric potential is the electric potential energy per unit charge. It means as the work required to move a positive test charge from a reference point to a specific point within an electric field.

$$\therefore V = -\int \vec{E}\cdot d\vec{l}$$



consider a 'Q' point at the origin.

$$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{a}_r$$

The potential difference between A and B

$$V_{AB} = -\int_{r_A}^{r_B} \vec{E}\cdot d\vec{l}.$$

$$d\vec{l} = dr.\hat{a}_r + r.d\theta.\hat{a}_\theta + r\sin\theta\, d\theta.\hat{a}_\phi$$

and, $\therefore \vec{E}\cdot d\vec{l} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}dr$

now,

$$V_{AB} = -\int_{r_A}^{r_B}\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}dr$$

$$= -\frac{Q}{4\pi\varepsilon_0}\int_{r_A}^{r_B}\frac{dr}{r^2}$$

$$= -\frac{Q}{4\pi\varepsilon_0}\left[-\frac{1}{r}\right]_{r_A}^{r_B} = \frac{Q}{4\pi\varepsilon_0}\left[r_B - r_A\right]$$

[Proved]

Ans. to the Q. NO-3

1st Part

An electric field is considered conservative if the work done in moving a charge between two points is independent of the path taken.

now,

Considering moving a unit test charge from A to B and back to A in different path.

$$\therefore V_{AB} = -V_{BA}$$

$$\therefore V_{AB} + V_{BA} = 0$$

So, The total work done over a closed path is zero,

$$\therefore \oint E \cdot dl = 0$$

This zero-sum results proves that the electric field E is conservative. [Showed]

2nd Part

$\vec{E}$ is conservative,

$$V_{AB} = -V_{BA}$$

$$\Rightarrow V_{AB} + V_{BA} = 0$$

$$\Rightarrow \oint_L \vec{E} \cdot \vec{dl} = 0$$

$$\Rightarrow \int_S (\vec{\nabla} \times \vec{E}) ds = 0 \quad \text{[Stokes theorem]}$$

$$\therefore \vec{\nabla} \times \vec{E} = 0$$

$\therefore$ Maxwell's 2nd equation $\vec{\nabla} \times \vec{E} = 0$

[Proved]

Ans. to the Q. No-4

An electric dipole consists of two equal and opposite Point charge separated by a small distanced. The electric diopole moment ($\vec{P}$) is a vector quantity defined as, $\vec{P} = Q\vec{d}$

$$V = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_1} - \frac{1}{r_2}\right] = \frac{Q}{4\pi\varepsilon_0}\left[\frac{r_2 - r_1}{r_1 r_2}\right]$$

if $r >> d$, $r_2 - r_1 \approx d\cos\theta$

$r_1 r_2 \approx r^2$

$$V = \frac{Q}{4\pi\varepsilon_0} \cdot \frac{d\cos\theta}{r^2} = \frac{Q\vec{d}\,\hat{a}_r}{4\pi\varepsilon_0 r^2} = \frac{\vec{P}}{4\pi\varepsilon_0 r^2}\hat{a}_r$$

[Proved]

$$\vec{E} = -\nabla V = -\left[\frac{\partial V}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\hat{a}_\phi\right]$$

$$\frac{\partial V}{\partial r} = \frac{\partial}{\partial r}\left(\frac{Q}{4\pi\varepsilon_0}\frac{d\cos\theta}{r^2}\right) = \frac{-Qd\cos\theta}{2\pi\varepsilon_0 r^3}$$

$$\frac{\partial V}{\partial \theta} = \frac{\partial}{\partial \theta}\left(\frac{Q}{4\pi\varepsilon_0}\frac{d\cos\theta}{r^2}\right) = \frac{-Qd\sin\theta}{4\pi\varepsilon_0 r^2}$$

$$\frac{\partial V}{\partial \phi} = 0$$

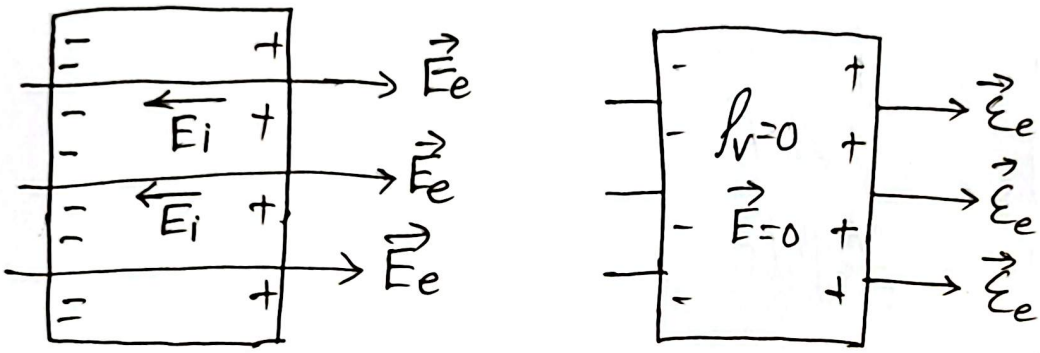
$$\therefore \vec{E} = \frac{Qd\cos\theta}{2\pi\varepsilon_0 r^3}\hat{a}_r + \frac{Qd\sin\theta}{4\pi\varepsilon_0 r^3}\hat{a}_\theta$$

$$= \frac{P}{4\pi\varepsilon_0}\left(2\cos\theta\,\hat{a}_r + \sin\theta\,\hat{a}_\theta\right)$$

[Proved]

Ans to the Q. No-05

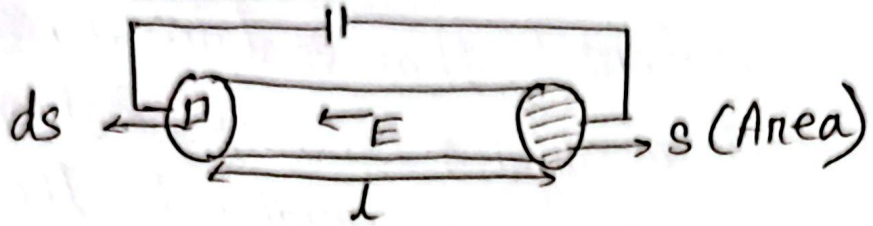
A conductor is a material that contains a high density of a free charge carries typically electrons that can move easily through the material under the influence of an electric field. Because of this mobality, conductors are excellent at transporting e^- energy and heat.



when external electric field is applied within the isolated conductor, the positive free charges are pushed along the same direction as applied field, while the negative charges move in the opposite direction. The free charges do two things. First, they accumulate on the surface of the conductor, and induced surface charge. Secondly, this surface charge setup an internal induced field $\vec{E}_i$, which cancels the externally applied field $\vec{E}_e$. So A perfect conductor ($\sigma = \infty$) can not contains an electrostatic field within it.

$$\therefore \vec{E} = -\nabla V = 0.$$

Ans to the Q. No-6



we know,

$$I = \int_s \vec{J}\, \vec{ds}$$

$$\Rightarrow I = \int_s \sigma \vec{E} \cdot \vec{ds}$$

$$\Rightarrow I = \sigma \int E \cdot \vec{ds}$$

$$\Rightarrow I = \sigma \int \frac{V}{L} \vec{ds}$$

$$\Rightarrow I = \frac{\sigma V}{L} \int ds$$

$$\Rightarrow I = \frac{\sigma V S}{L}$$

$$\Rightarrow \frac{I}{V} = \frac{\cancel{\sigma S}}{\cancel{L}} \frac{L}{\sigma S}$$

$$\therefore R = \frac{L}{\sigma S}$$

$\boxed{\text{Proved}}$

Ans. to the Q. No-7

Dielectric: A dielectric is an insulating material that doesn't conduct electricity but can store electrical energy. when placed in an electric field, it allows the field to pass through it while shifting it's internal charges.

i) **Polar dielectric:** A polar dielectric is a material whose molcules have a permanent electric dipole moment even without an external electric field.

ii) **Nonpolar dielectric:** A non polar dielectric is a material whose molecules do not have a permanent dipole moment but it become polarized when an external electric field is applied.

iii) **Linear dielectric:** A dielectric material where the induced polarization (P) is directly proportional to the strength of applied electric field (E)
($P \propto E$)

iv) **Isotropic Dielectric:** An isotropic dielectric has the same electrical properties in all direction. It's behavior does not depend on direction.

vi) Homogeneous Dielectric: A homogeneous dielectric has uniform composition and properties throughout the material. Its dielectric properties are the same at every point.

vi) Dielectric strength: The maximum electric field that a dielectric material can withstand without breaking down and becoming a conductor. It is a measure of how good an insulator the material is.

vii) Dielectric Breakdown: Dielectric breakdown is the process where an insulating material becomes electrically conductive when subjected to an electric field exceeding its maximum threshold.